



Hochschule München

University of Applied Sciences

Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Heykel Khadhraoui

Computational Engineering (M.Sc.)

Supervisors: Prof. Norbert Nitzsche · Prof. Ruth Otto

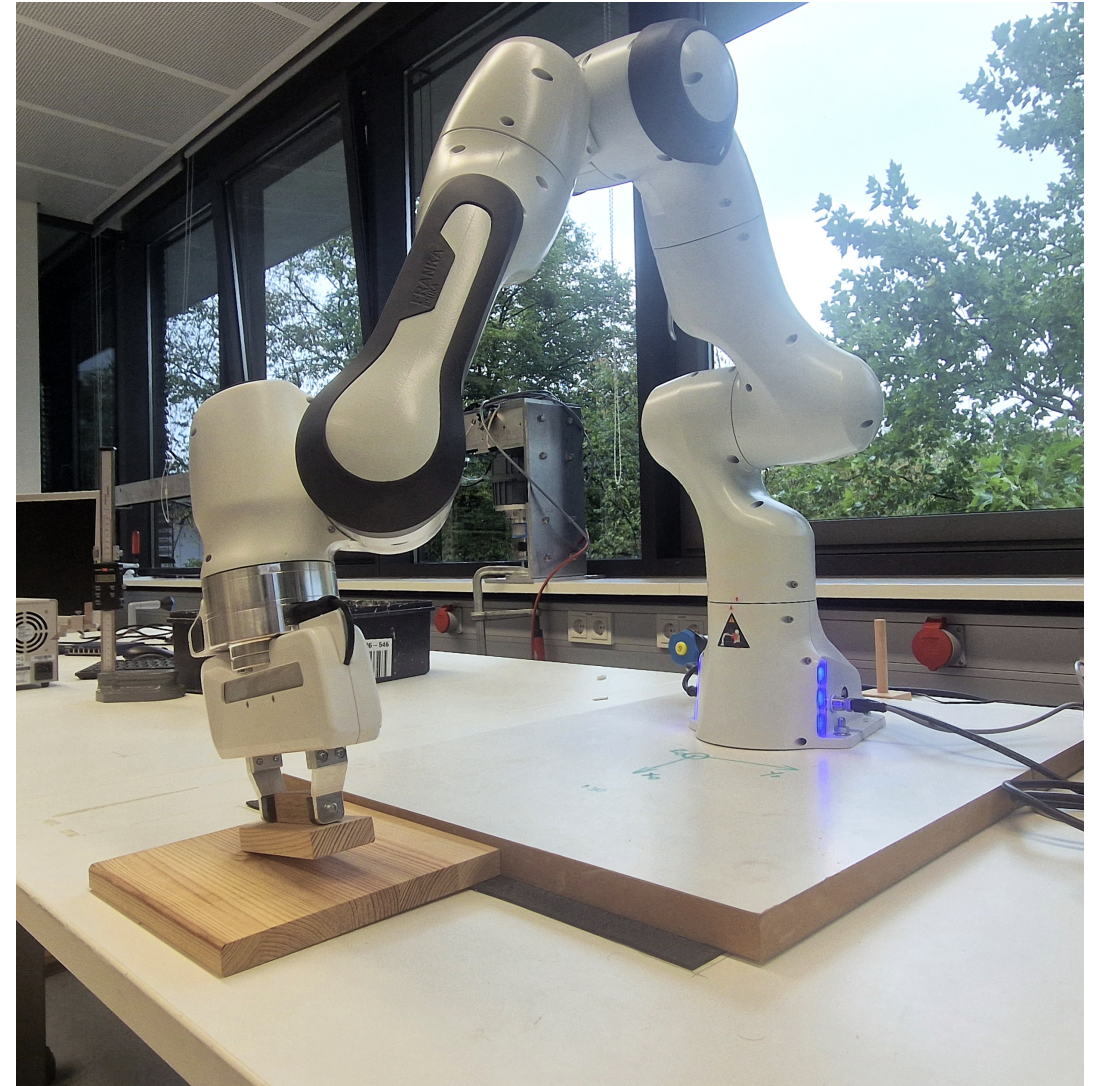
Date: 30.09.2026

- Problem

- Grinding requires an aligned tool face.
- Calibration leaves uncertainty in the physical surface orientation.
- Joint friction can prevent the desired tool alignment with the configured surface.

- Idea

- Cartesian impedance makes position and rotation compliant.
- Pressing a tilted tool creates a contact moment.
- Rotational compliance allows the tool to turn towards alignment.

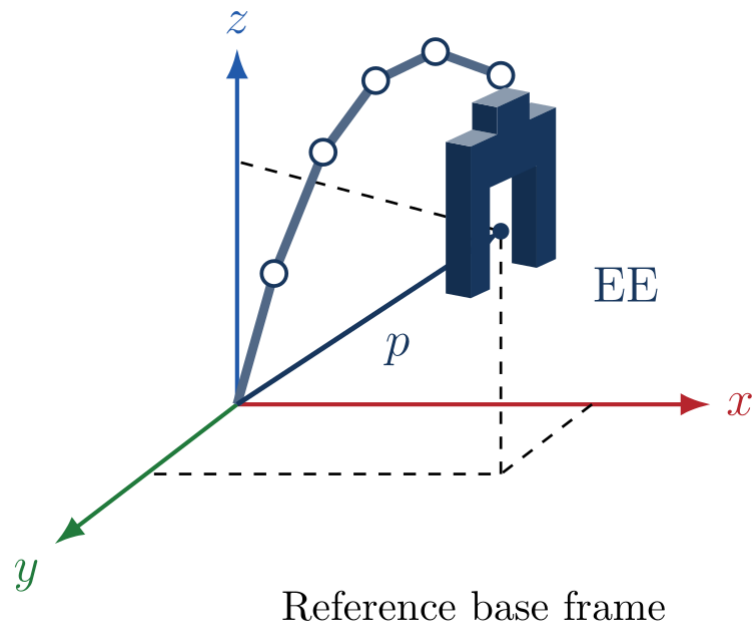


Contact demonstration

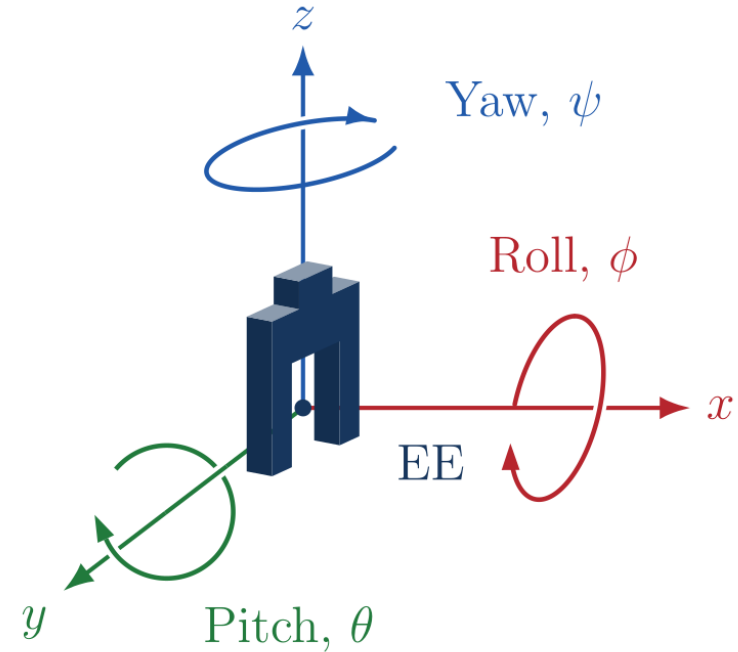


1. Introduction
2. Cartesian controller
3. Contact experiments
4. Contact results
5. Null-space control and experiments
6. Conclusion and future work

- Position: 3 translations

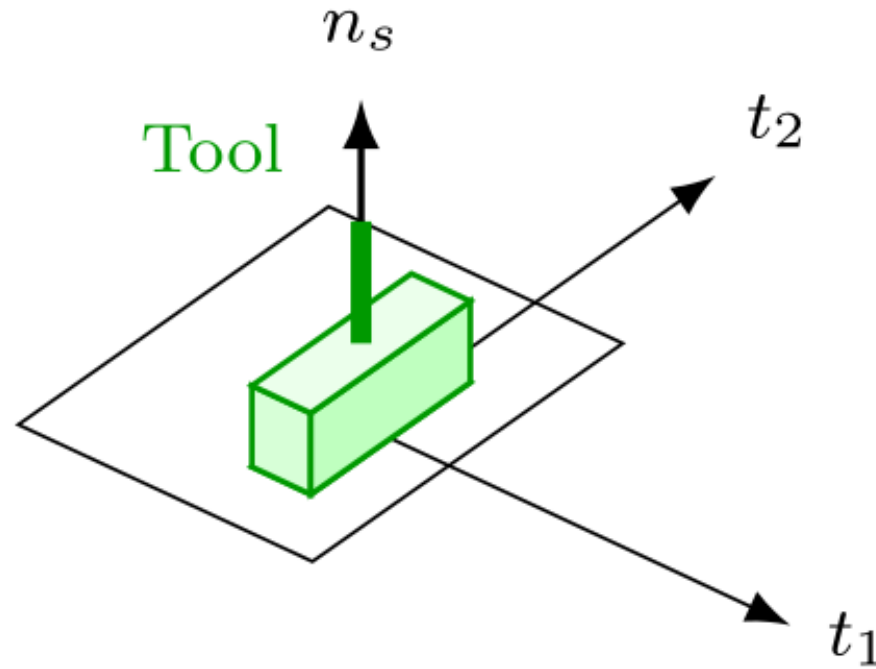


- Orientation: 3 rotations



- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

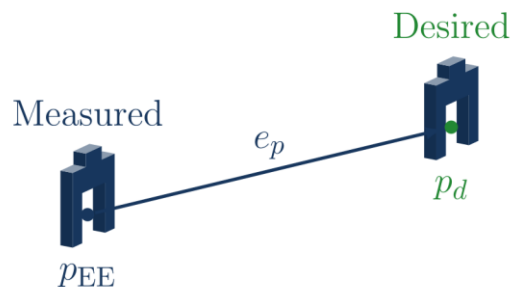
- Surface-relative compliance



n_s outward surface normal
 t_1, t_2 perpendicular surface tangents

- Positional error

$$e_p = p_d - p_{EE}$$



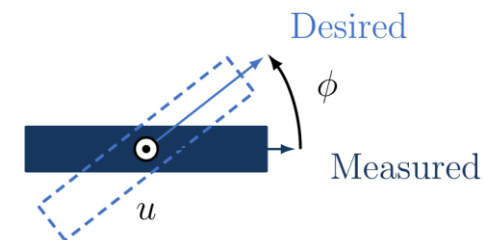
$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}}_{K_C} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}}_{D_C} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

- Rotational error

$$e_R = \phi u$$



$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

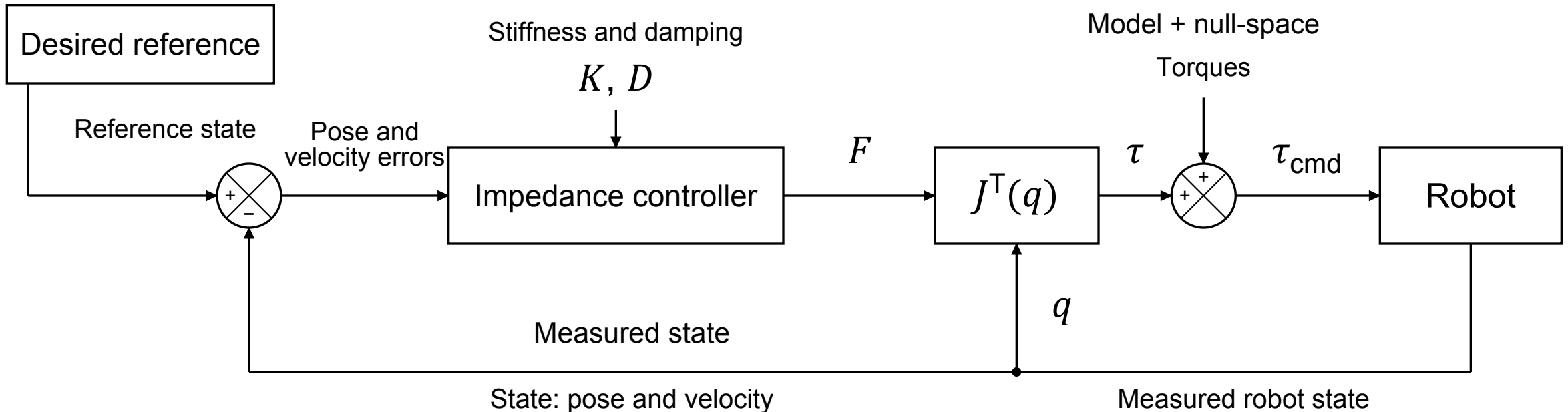
- In the surface frame, K and D are diagonal and decouple translation and rotation.

- Joint-torque mapping

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau = J^T(q)F$$

- Jacobian

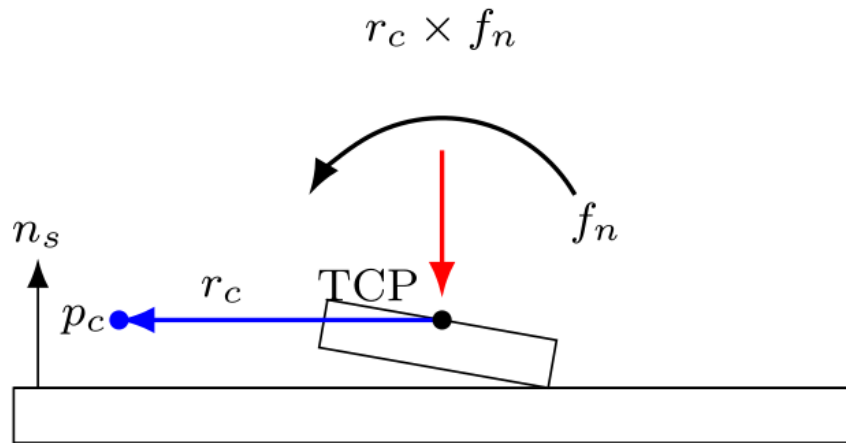
$$\begin{bmatrix} \dot{p}_{EE} \\ \omega_{EE} \end{bmatrix} = J(q)\dot{q}$$



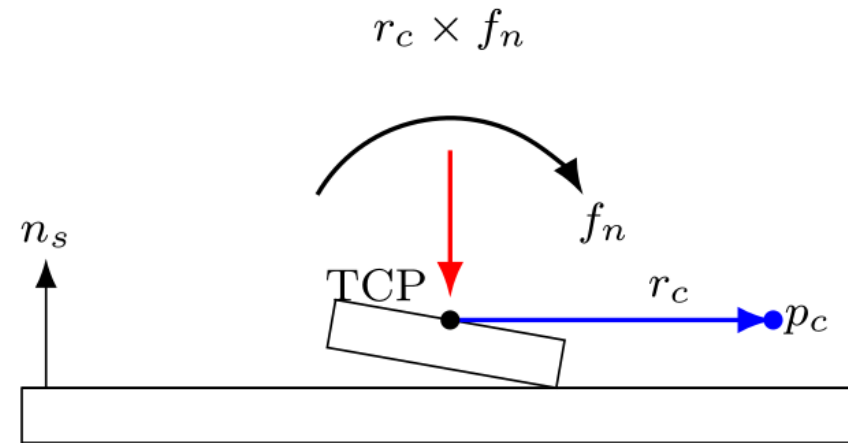
Real-time control with a 1 ms cycle

Centre of compliance (CoC)

- Effect of CoC displacement
- Added moment can support or oppose rotation.



(a) Supporting moment



(b) Opposing moment

- Shifting the impedance reference point

Tool centre point (TCP): controlled reference point on the EE

p_c Virtual centre of compliance (CoC): new reference

- Choosing an impedance reference point away from the TCP couples force and moment.
- The commanded force acts as if applied at the virtual point p_c .

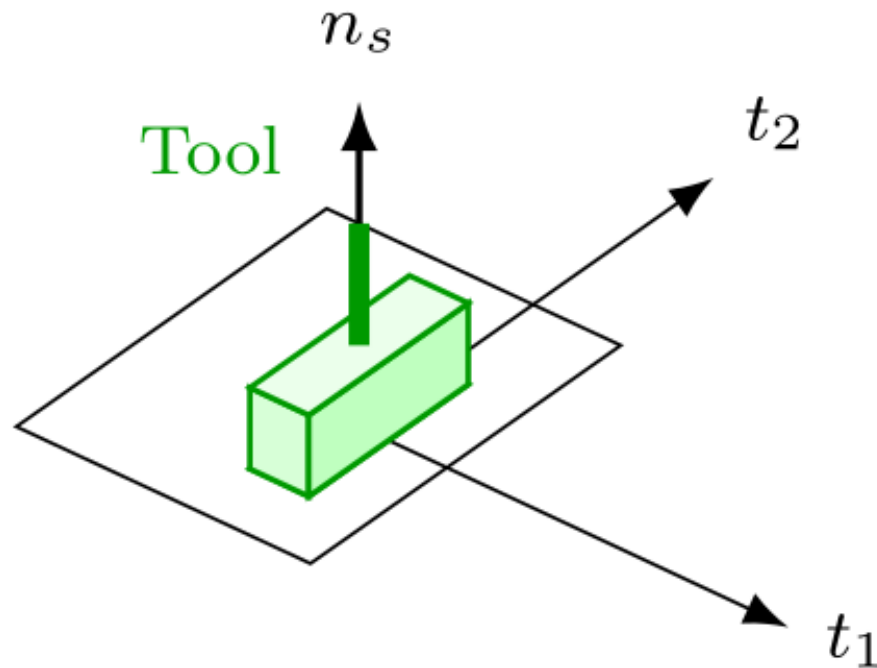
$r_c = p_c - p_{\text{TCP}}$ Displacement from TCP to CoC

$A = \text{Ad}(r_c)$ Point-shift adjoint

$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$

- Non-zero off-diagonal blocks in K and D couple translation and rotation.

Tool orientation → Surface approach → Contact establishment → Surface grinding



- Compliant contact settings

- Normal translation

$$K_{p,n} = 350 \text{ N/m}$$

- Tangential translation

$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

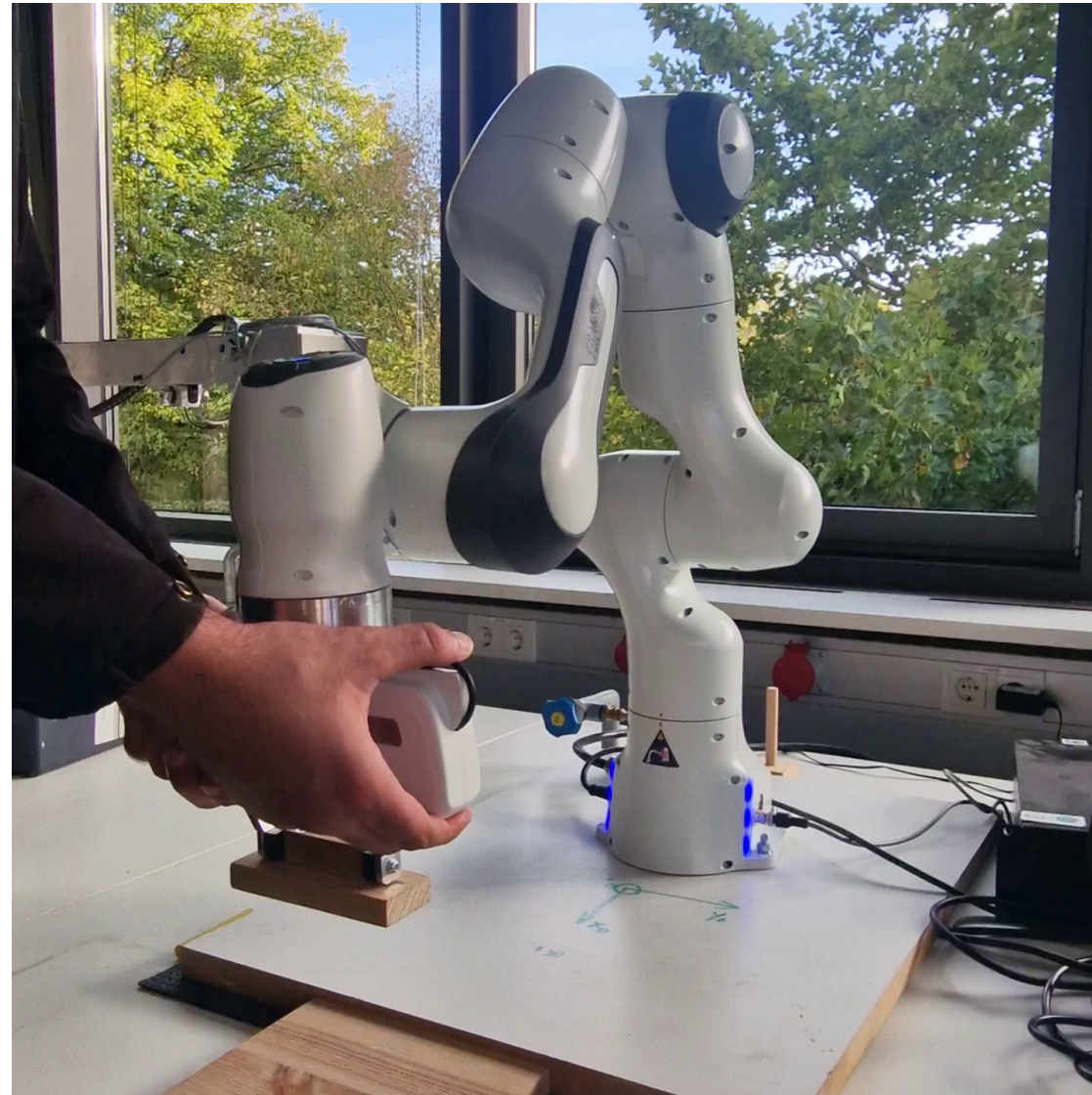
- Rotation about the surface normal

$$K_{R,n} = 50 \text{ N m/rad}$$

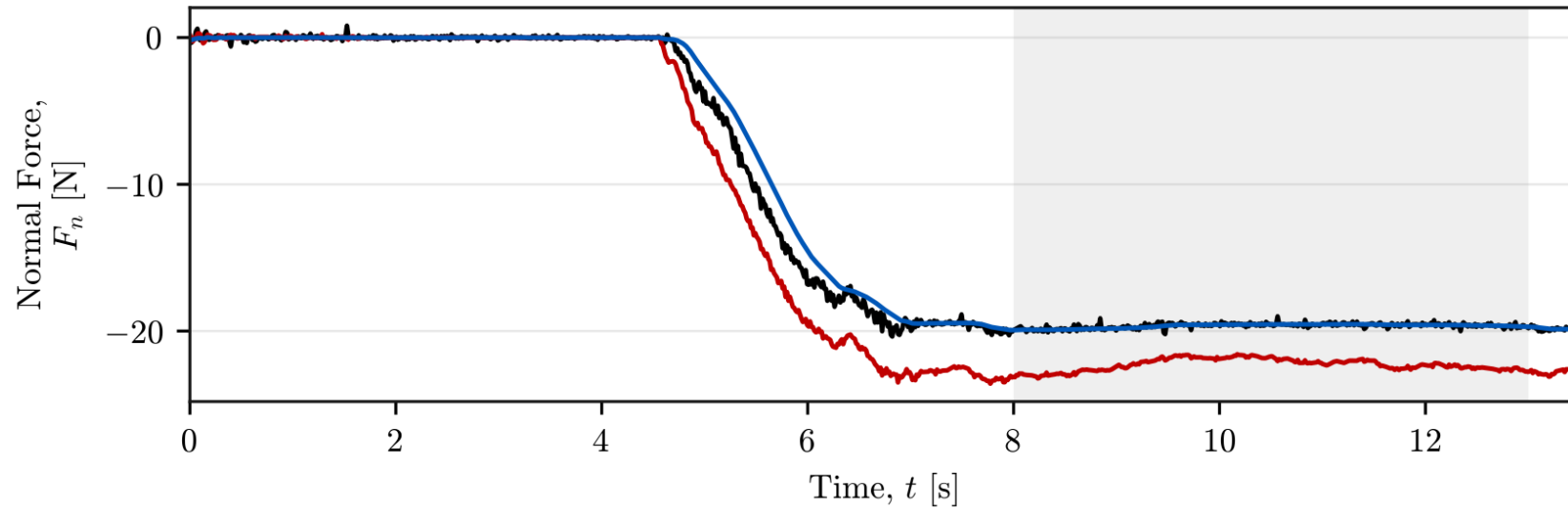
- Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$

Plausibility experiment



Normal-force plausibility assessment



— Commanded

— Model-estimated

— Quasi-static prediction

Manual push: about 20 mm

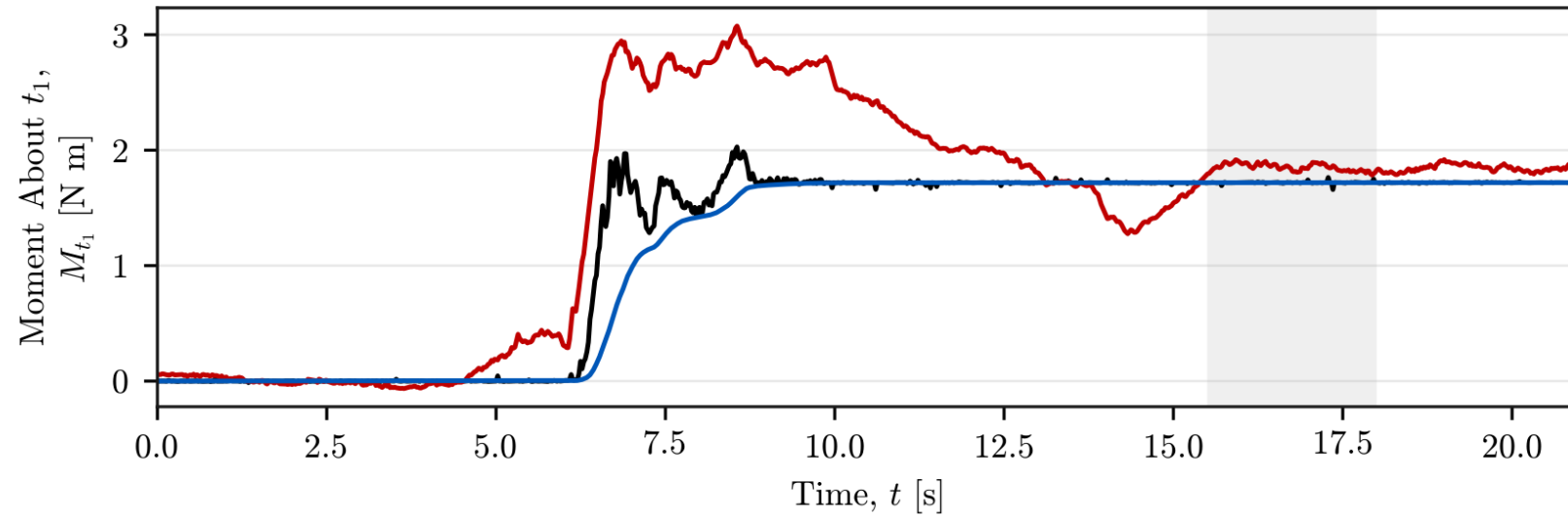
$$\begin{aligned} F_{n,qs} &= K_{p,n} n_s^\top (e_p - e_{p,0}) \\ &= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N} \end{aligned}$$

Quasi-static prediction
-19.65 N

Commanded mean
-19.64 N

Model-estimated mean
-22.29 N

Moment plausibility assessment



— Commanded

— Model-estimated

— Quasi-static prediction

Manual rotation: about 6.5°

$$\begin{aligned} M_{t_1,qs} &= K_{R,t_1} t_1^\top (e_R - e_{R,0}) \\ &= (15 \text{ N.m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N.m} \end{aligned}$$

Quasi-static prediction

1.72 N.m

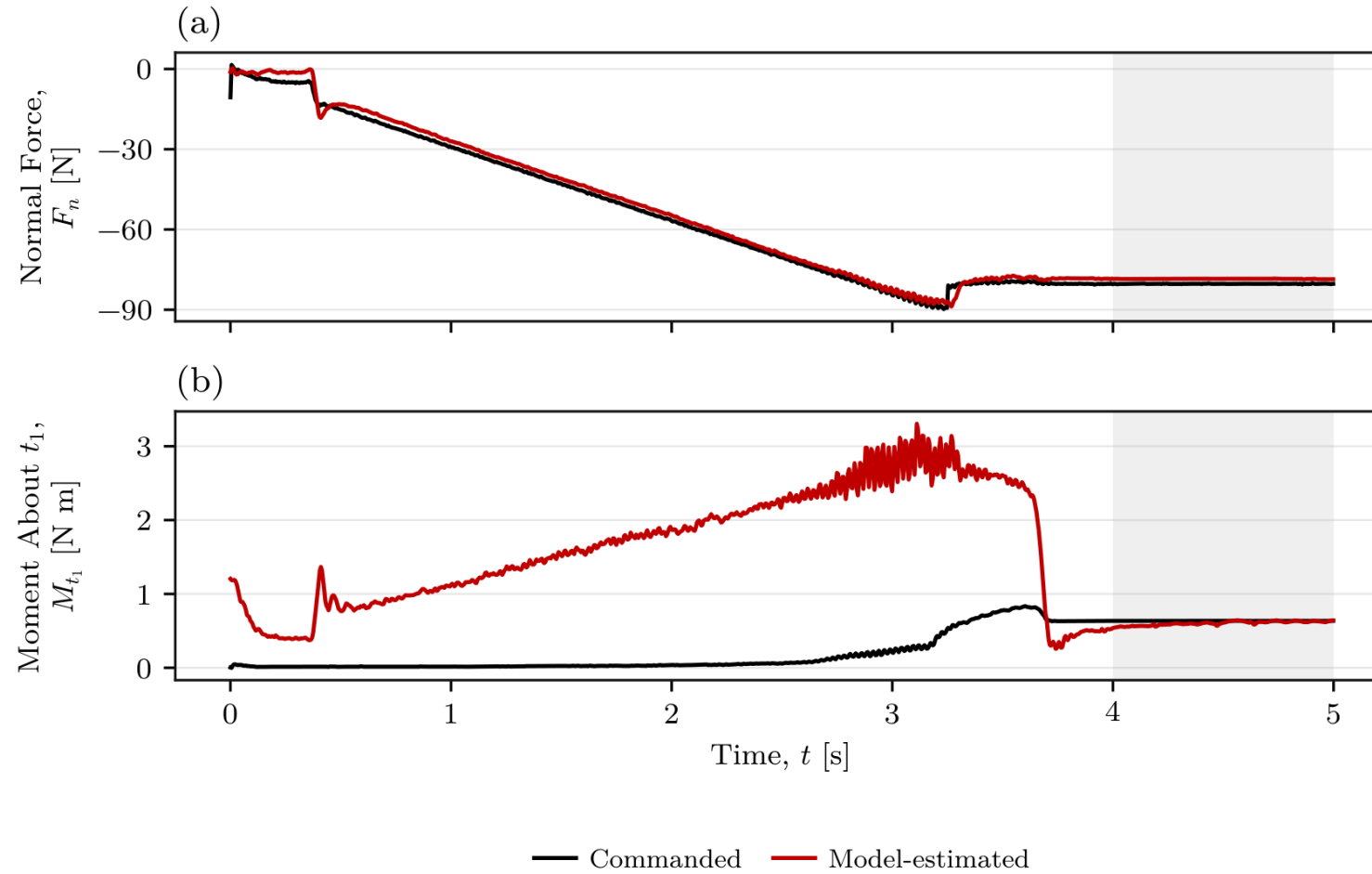
Commanded mean

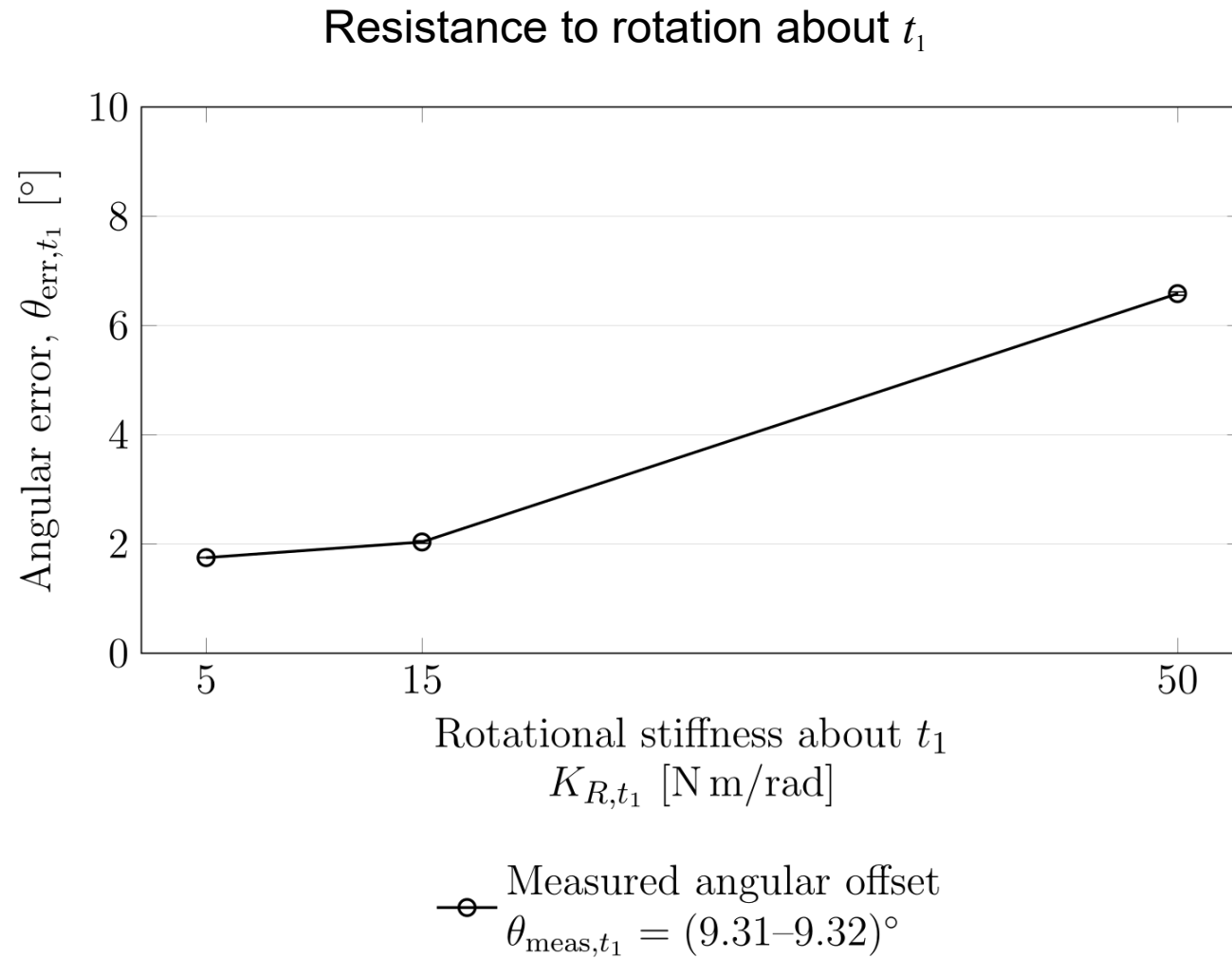
1.72 N.m

Model-estimated mean

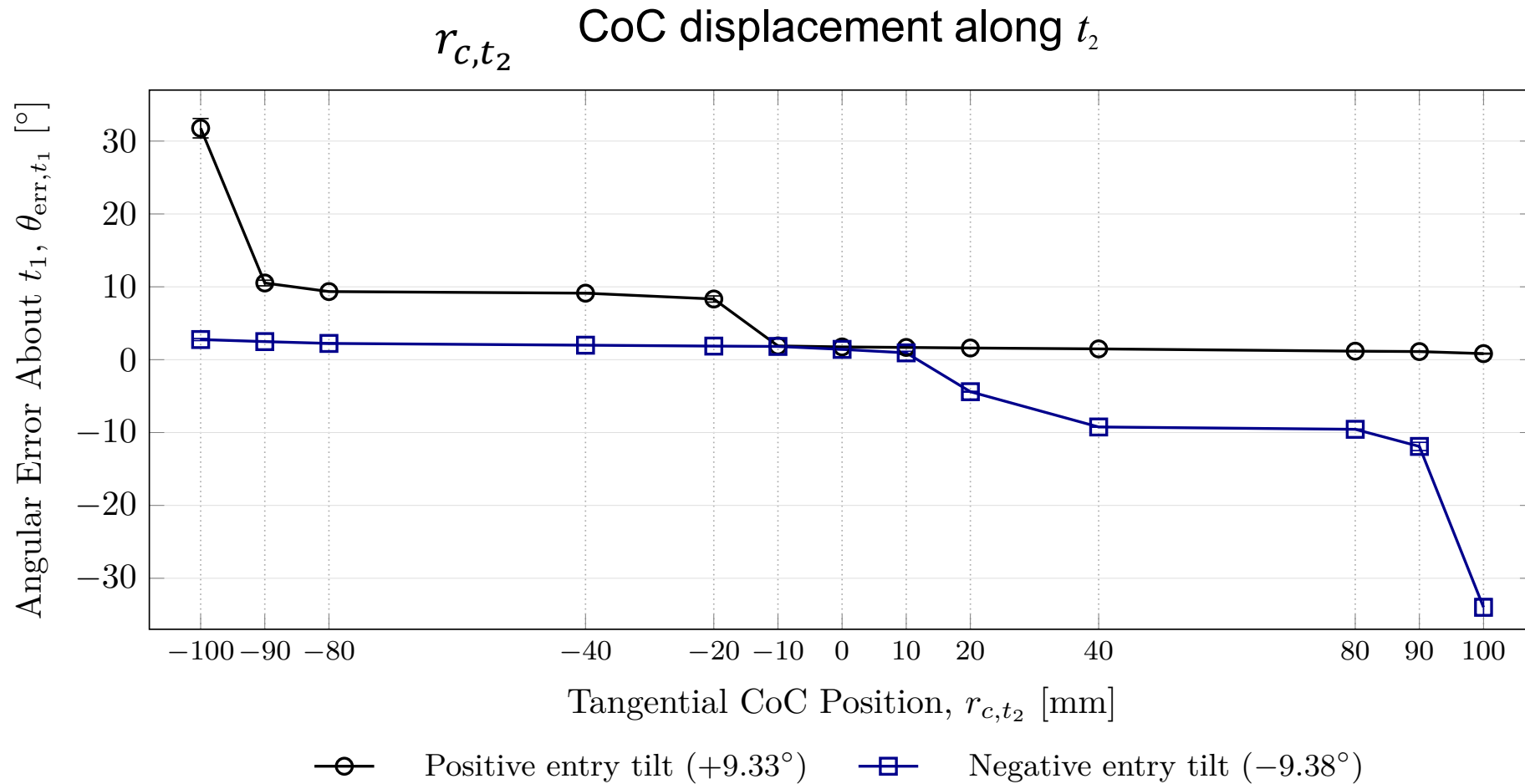
1.85 N.m

Commanded and estimated wrench

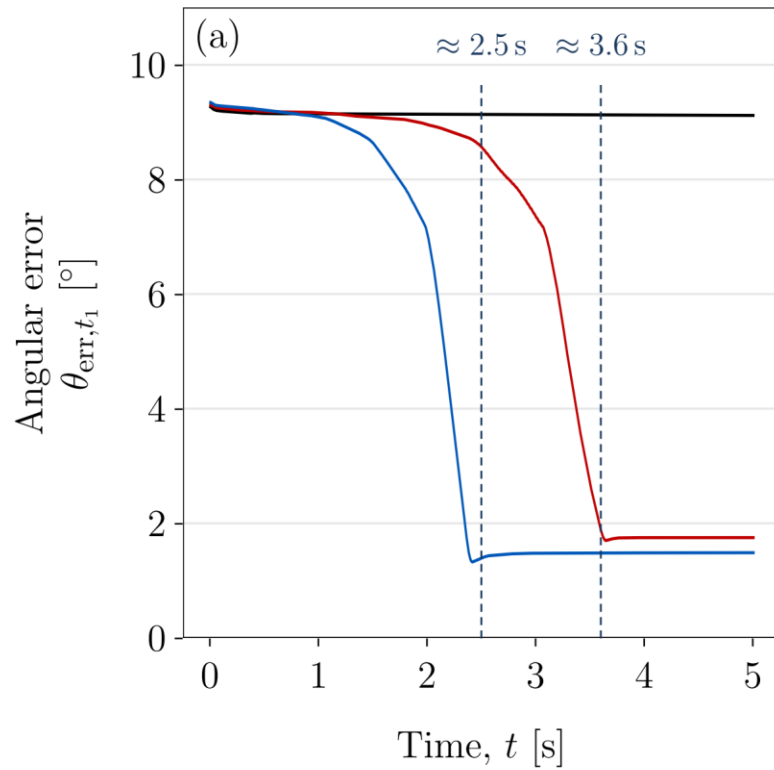




Effect of CoC position

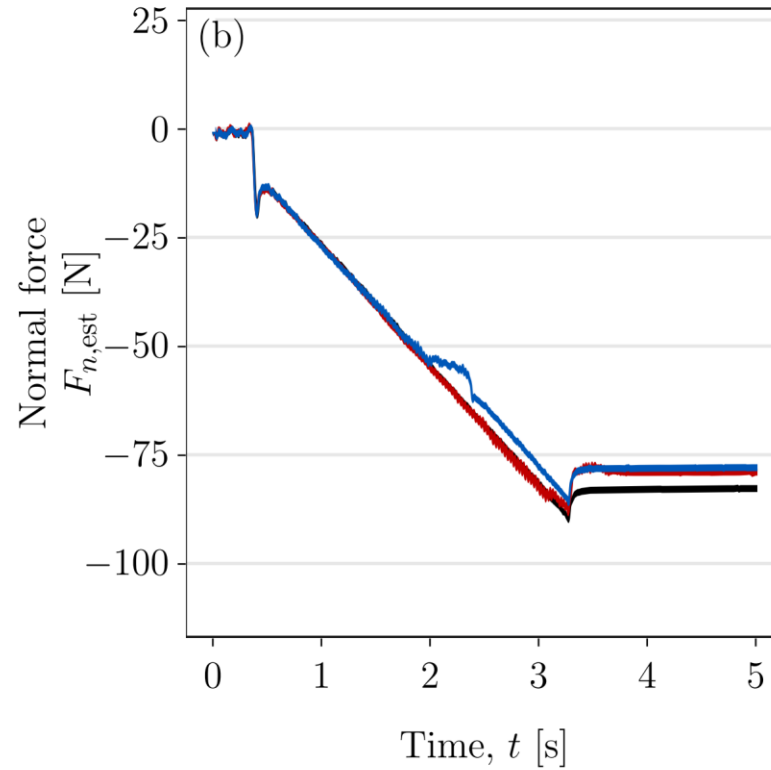


- Angular error



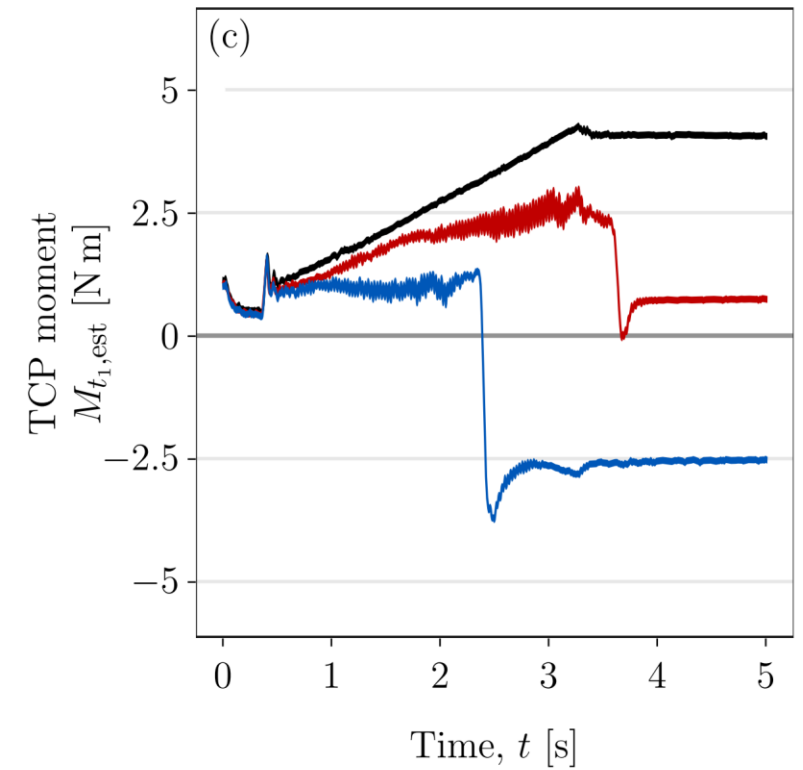
— $r_{c,t_2} = -40\text{ mm}$

- Normal force



— TCP: $r_{c,t_2} = 0$

- TCP moment



— $r_{c,t_2} = +40\text{ mm}$

Redundant direction:

- One null-space direction at full Jacobian rank.
- Several joints move together.

$$\dim(\ker J) = 7 - 6 = 1$$

Damping torque τ_d :

- Opposes and damps null-space velocity.

Primary Cartesian task:

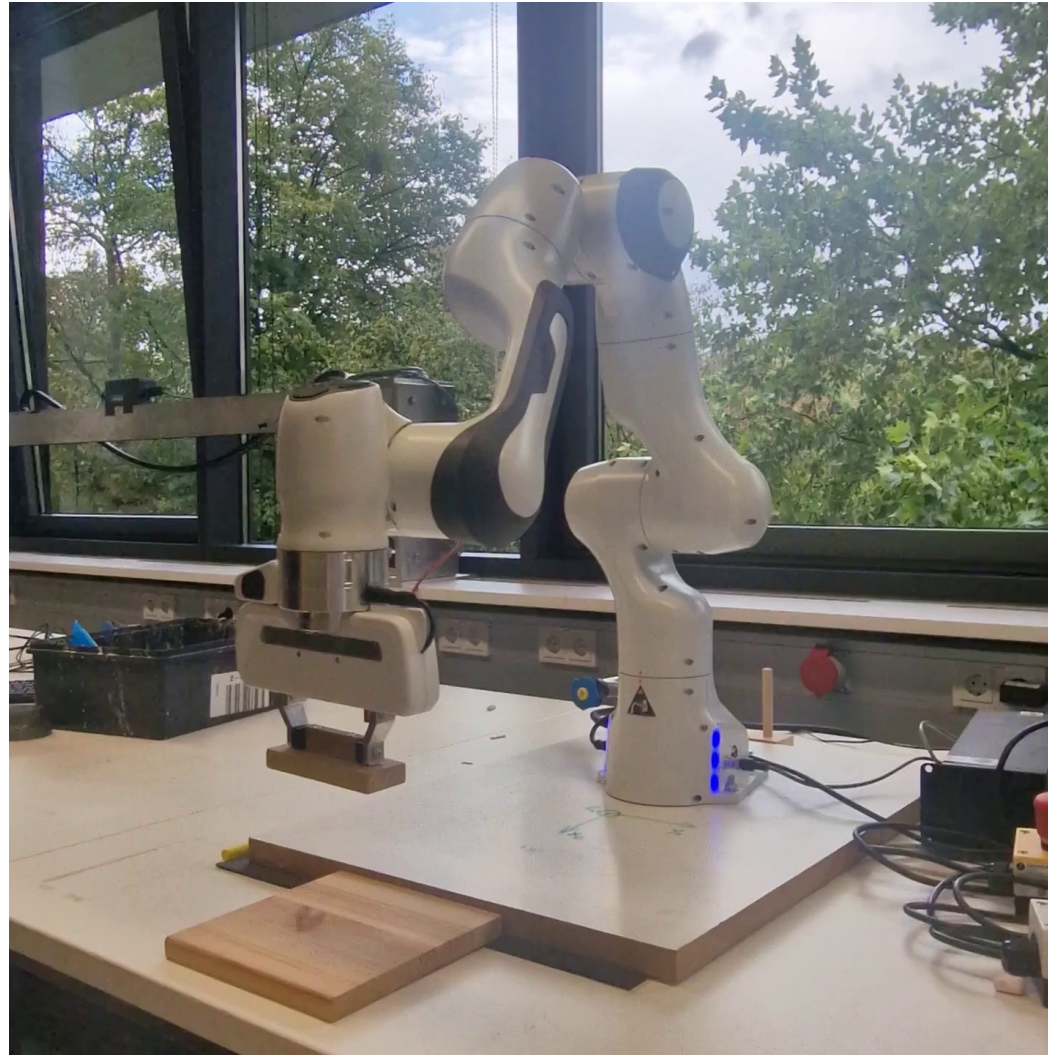
- Null-space motion preserves the instantaneous motion of the EE.

$$J(q) \dot{q}_{\text{null}} = 0$$

Conditioning torque τ_σ :

- Adjusts the joint configuration to improve Jacobian conditioning.
- At a singularity, the EE loses a motion direction.
- Near it, some EE motions require very large joint velocities.

Disturbance demonstration



Demonstration · Part 1

Disturbance demonstration



Demonstration · Part 2

- Commanded joint disturbance

$$\tau_{\text{dist}} = J(q)^{\top} f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

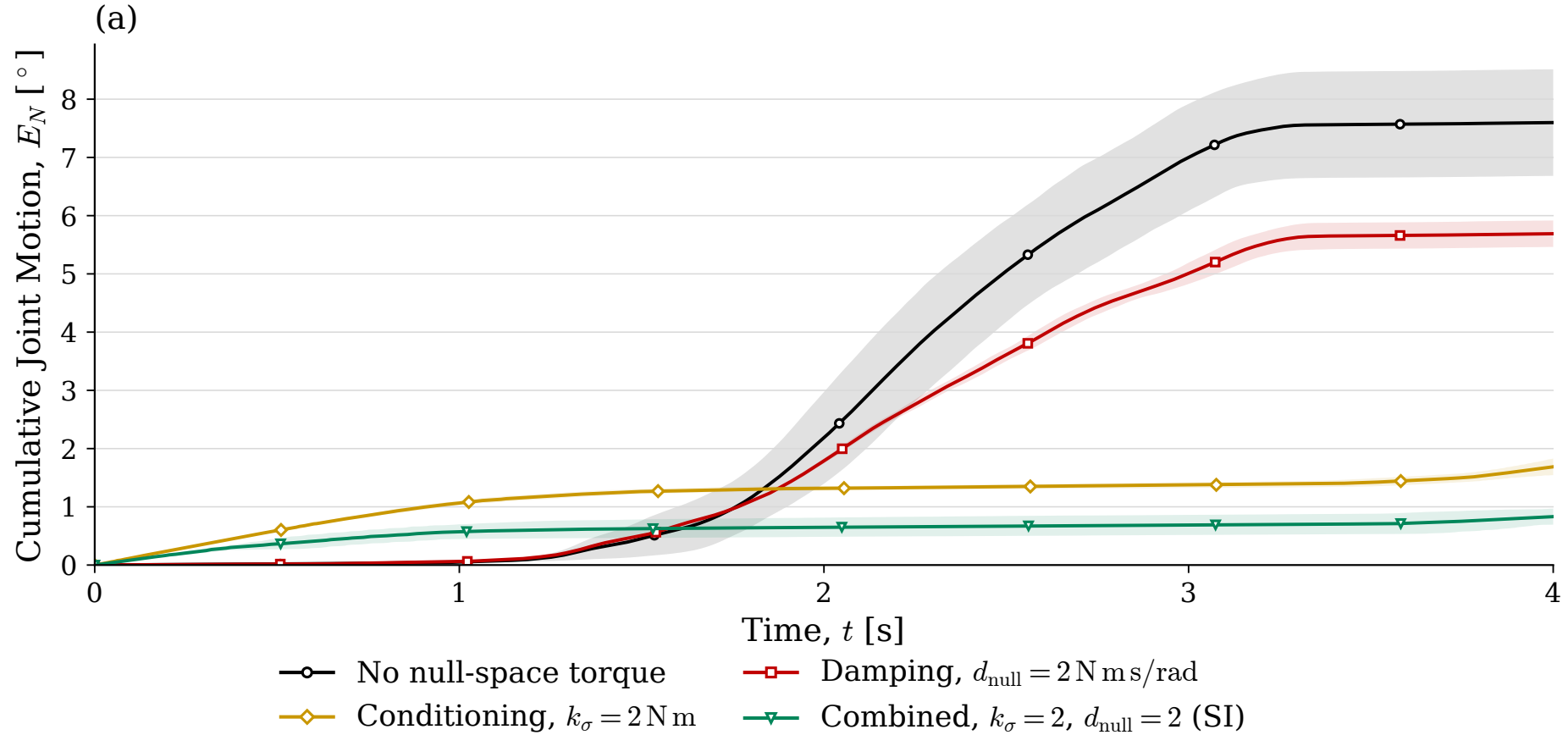
- Four conditions with three trials each
 - No null-space torque
 - Conditioning: $k_{\sigma} = 2 \text{ N m}$

- Cartesian pose hold

Hold the captured position and orientation.

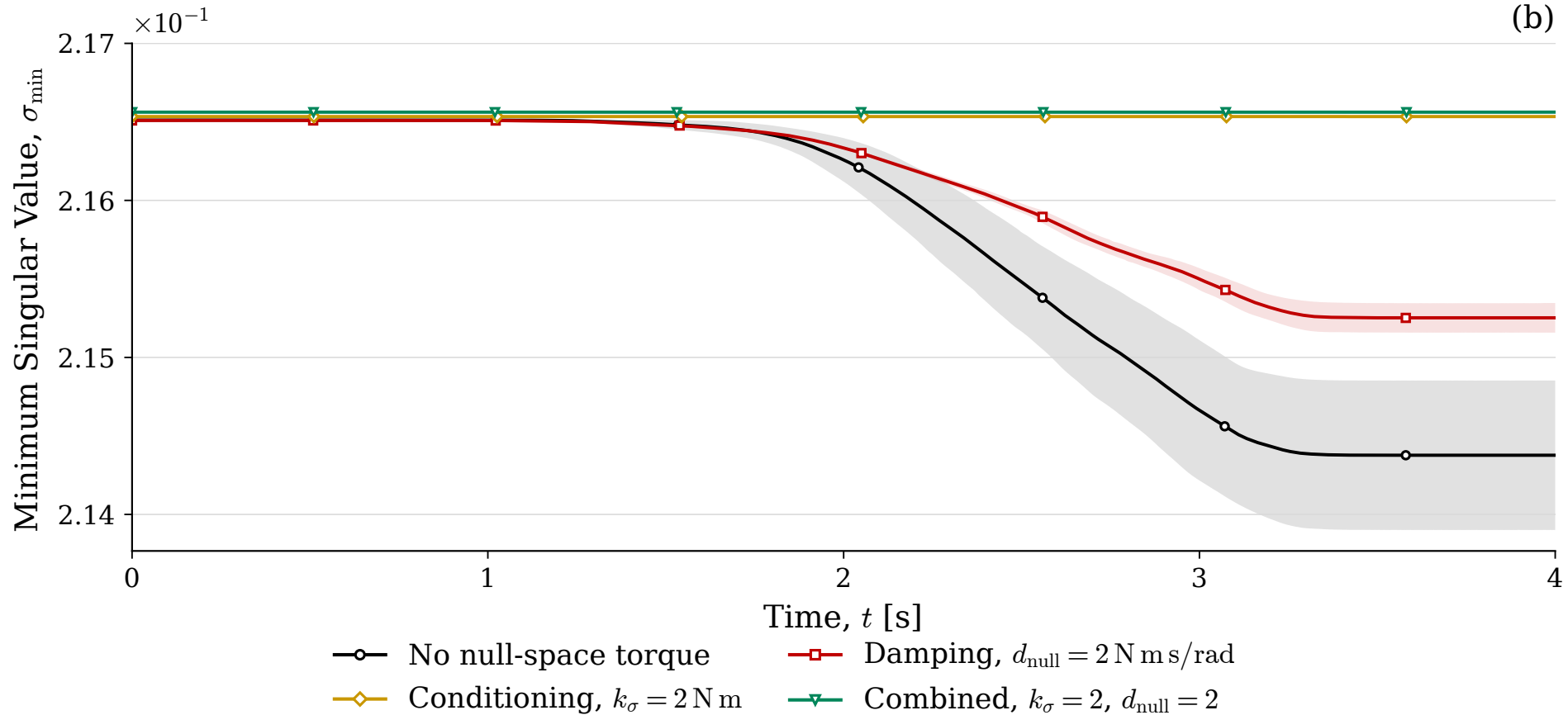
- Damping: $d_{\text{null}} = 2 \text{ N m s/rad}$
- Combined: conditioning + damping
 $k_{\sigma} = 2 \text{ N m}$ and $d_{\text{null}} = 2 \text{ N m s/rad}$

Cumulative joint motion



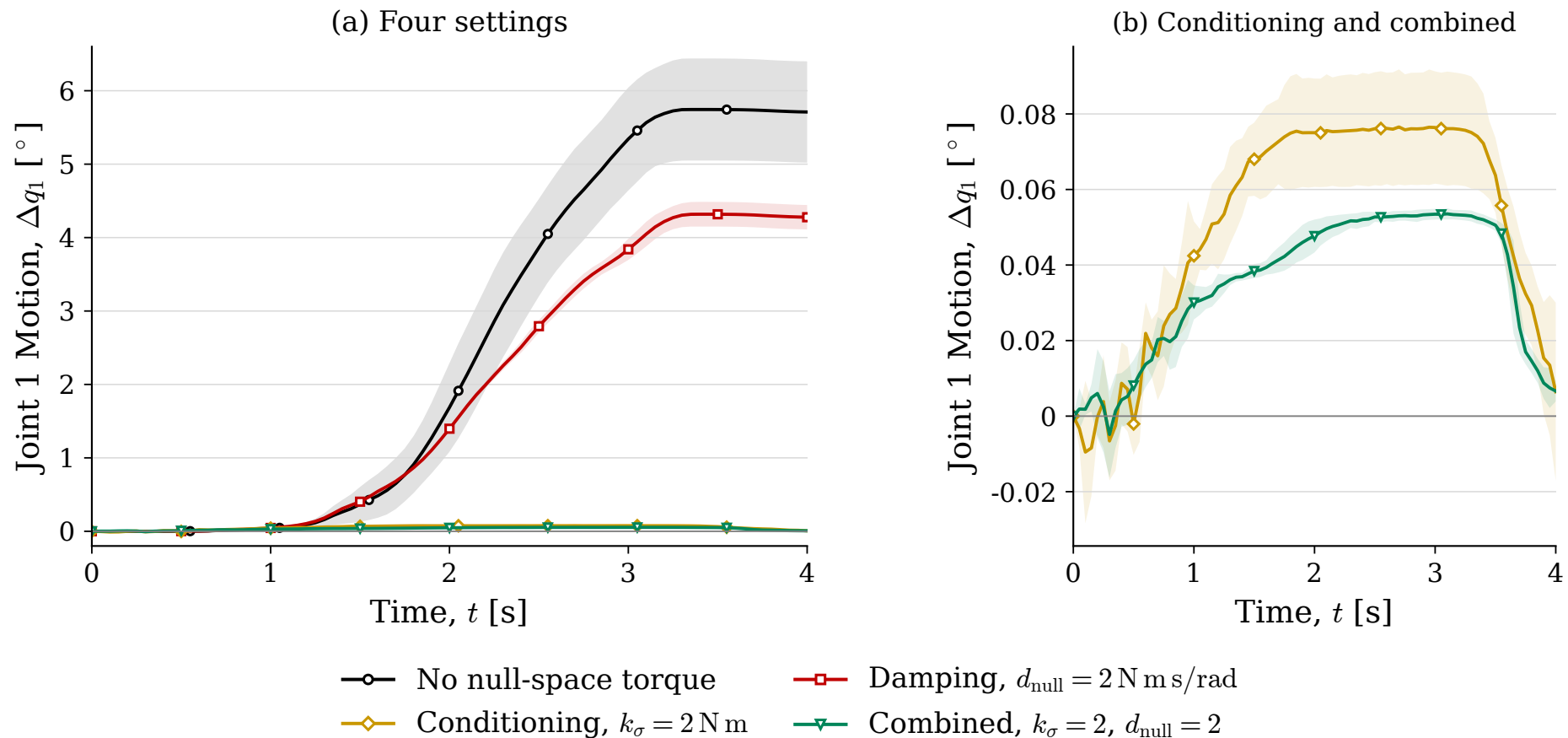
Three-trial mean and one sample standard deviation.

- Damping alone reduced cumulative motion by 25.1% compared with no torque.
- At 2 N m, adding damping to conditioning reduced cumulative motion by 50.7%.



- The indicator decreased with no torque and damping alone.
- Conditioning and combined control kept the indicator nearly constant.

Joint motion over time



Three-trial mean and one sample standard deviation.

- Conditioning at 2 N m shows repeated joint-motion reversals.
- Adding damping reduces total motion, while reversals remain.

- Compliant rotation allowed contact moments to reduce angular error
- Higher rotational stiffness left a larger angular error
- The CoC could support or oppose the contact moment
- Null-space control reduced redundant motion under disturbance
- Future work
 - Use a more rigid tool mount for better measurements
 - Adapt the CoC to support the contact moment, then return it to the TCP after alignment

Opposing moment



- Null-space projector

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ EE velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

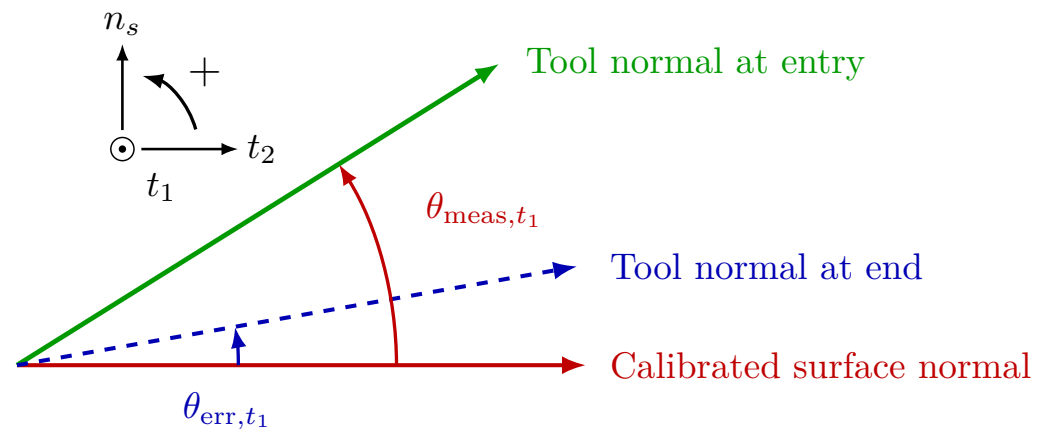
$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning.
Can act at zero joint velocity.

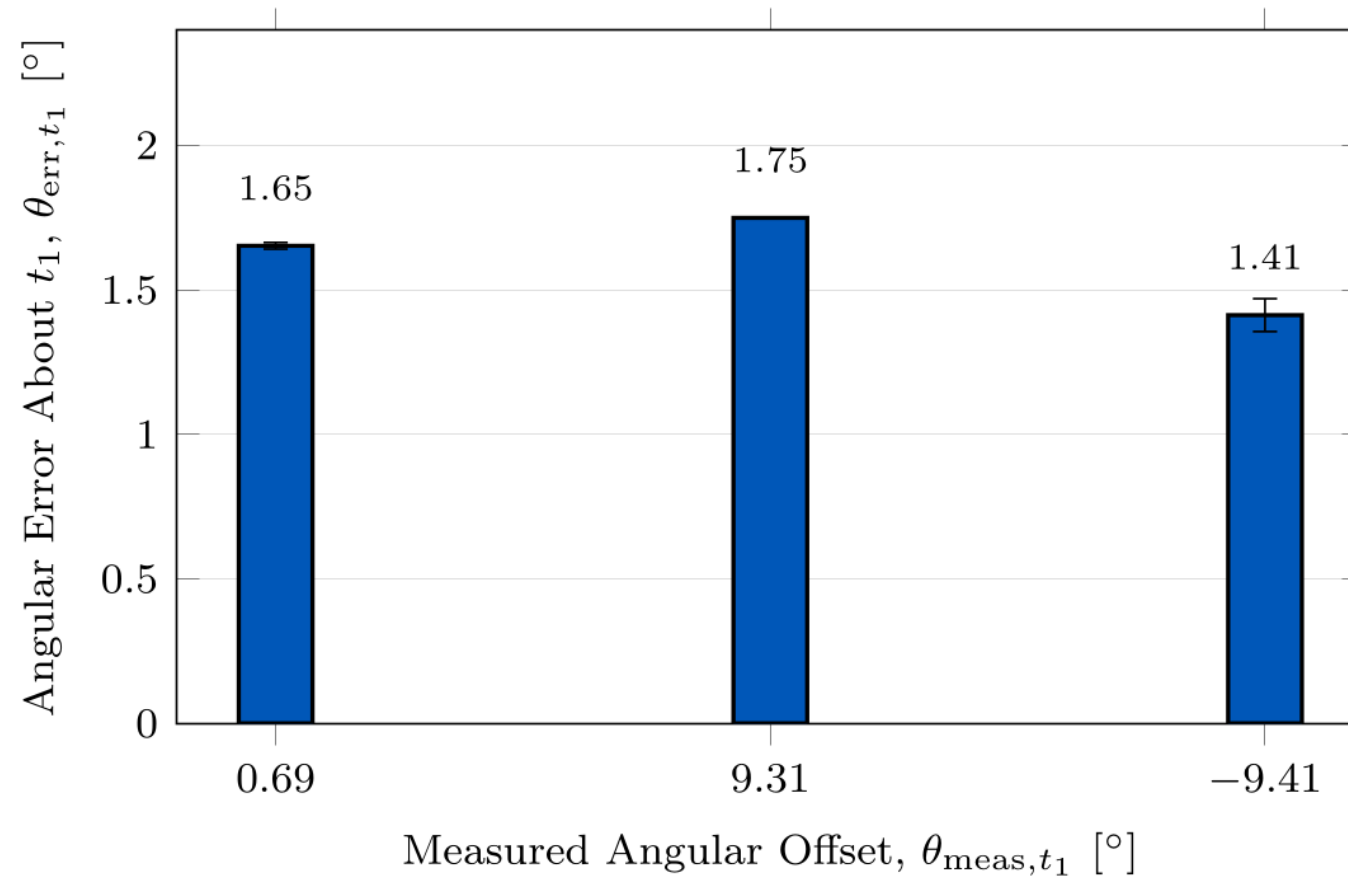
k_σ : torque magnitude [N m]

v_7 : unit null-space direction

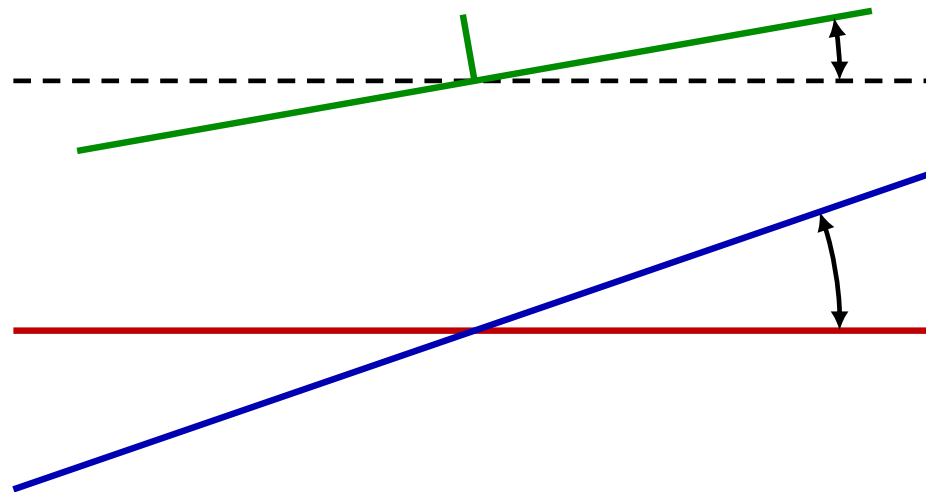
s_σ : direction choice, zero in deadband



Baseline angular error



Sources of angular offset



Desired–achieved difference

Configured–physical difference

— Configured surface

- - - Desired orientation

— Physical surface

— Tool face